

Interest of ZAP-70 in the prognostic evaluation of Chronic Lymphocytic Leukemia

K Bouzenda ⁽¹⁻³⁾, R Gourmat⁽²⁾, A Zidani⁽²⁾, E Mahdjoub⁽¹⁻³⁾, I Hammoudi⁽¹⁾, S Kebaili⁽³⁾, Z Ouchenane ⁽³⁾.

1-Hemobiology-Blood Transfusion Department, Constantine University Hospital Center, Constantine, Algeria.
2- Department of Pharmacy, Faculty of Medicine, Constantine3 University. Constantine, Algeria.
3- Laboratory of Biology and Molecular Genetics, Faculty of Medicine, Constantine3 University, Constantine, Algeria
3- Hematology Department, Constantine University Hospital Center, Constantine, Algeria.



OBJECTIVES

Chronic lymphocytic leukemia is a malignant hematological disorder with heterogeneous progression. The ZAP-70 protein (Zeta-chain-associated protein kinase 70), normally expressed in T lymphocytes, constitutes a biological prognostic marker whose clinical value in predicting the progression of chronic lymphocytic leukemia has garnered increasing interest in recent years. This study aims primarily to evaluate the significance of ZAP-70 expression in the prognostic assessment of CLL as an effective and reliable parameter by comparing it to the Binet classification and the progression of the disease, in order to better assess patient prognosis and optimize management strategies from the time of diagnosis.

MATERIEL & METHODS

Sampling

This descriptive and analytical cross-sectional study focused on 39 patients with chronic lymphocytic leukemia (CLL), including cases primarily followed up in the hematology department of CHU Constantine, presenting different clinical and evolutionary stages of the disease.

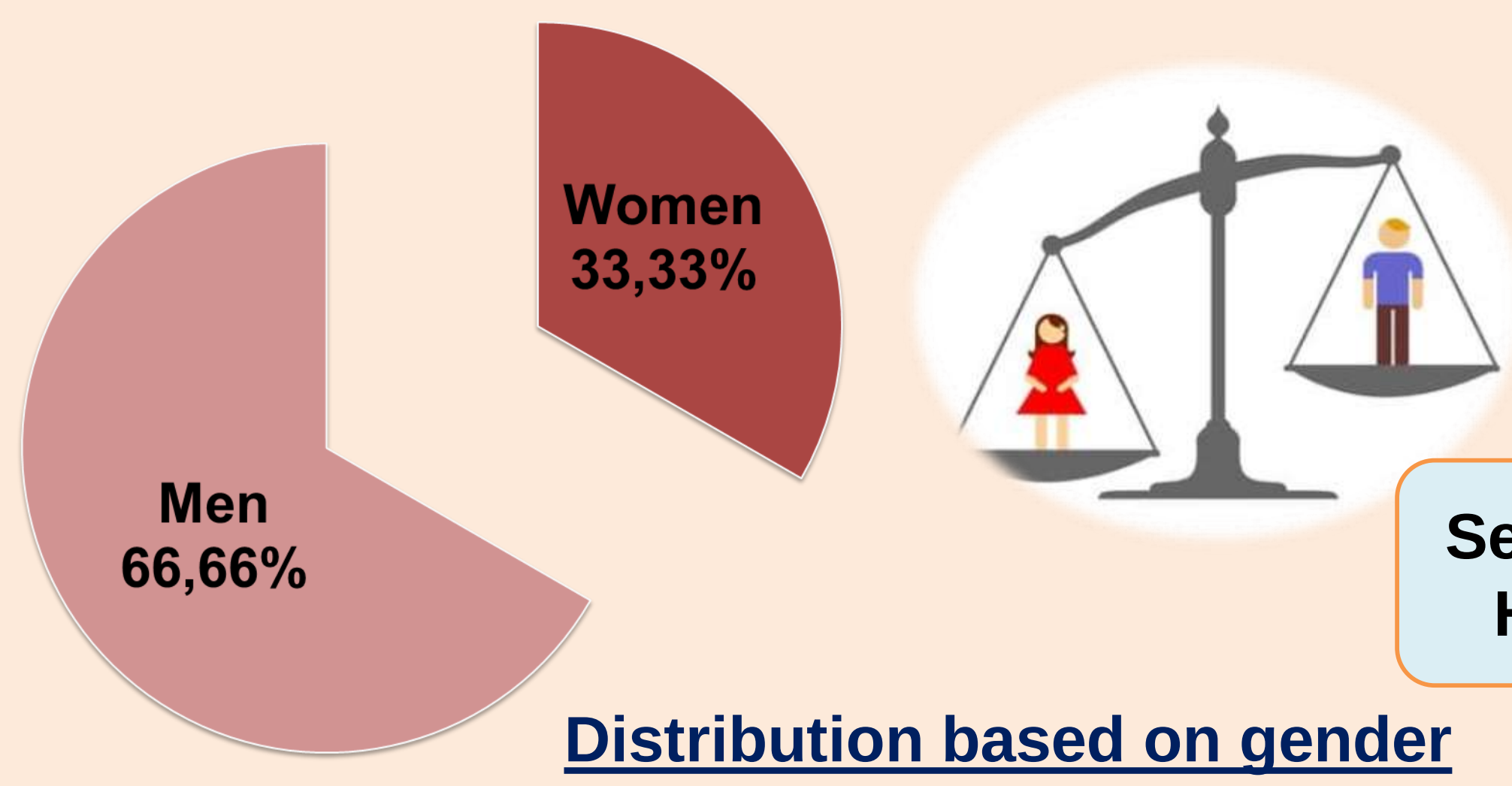
information gathering

For each patient, we conducted an assessment and evaluation of their clinical-biological progression, by monitoring the white blood cell and lymphocyte counts in particular, the hemoglobin and platelet levels, the onset of complications related to their pathology, the progression of the number of affected lymph node areas, the response to treatments, and the duration of survival without treatment.

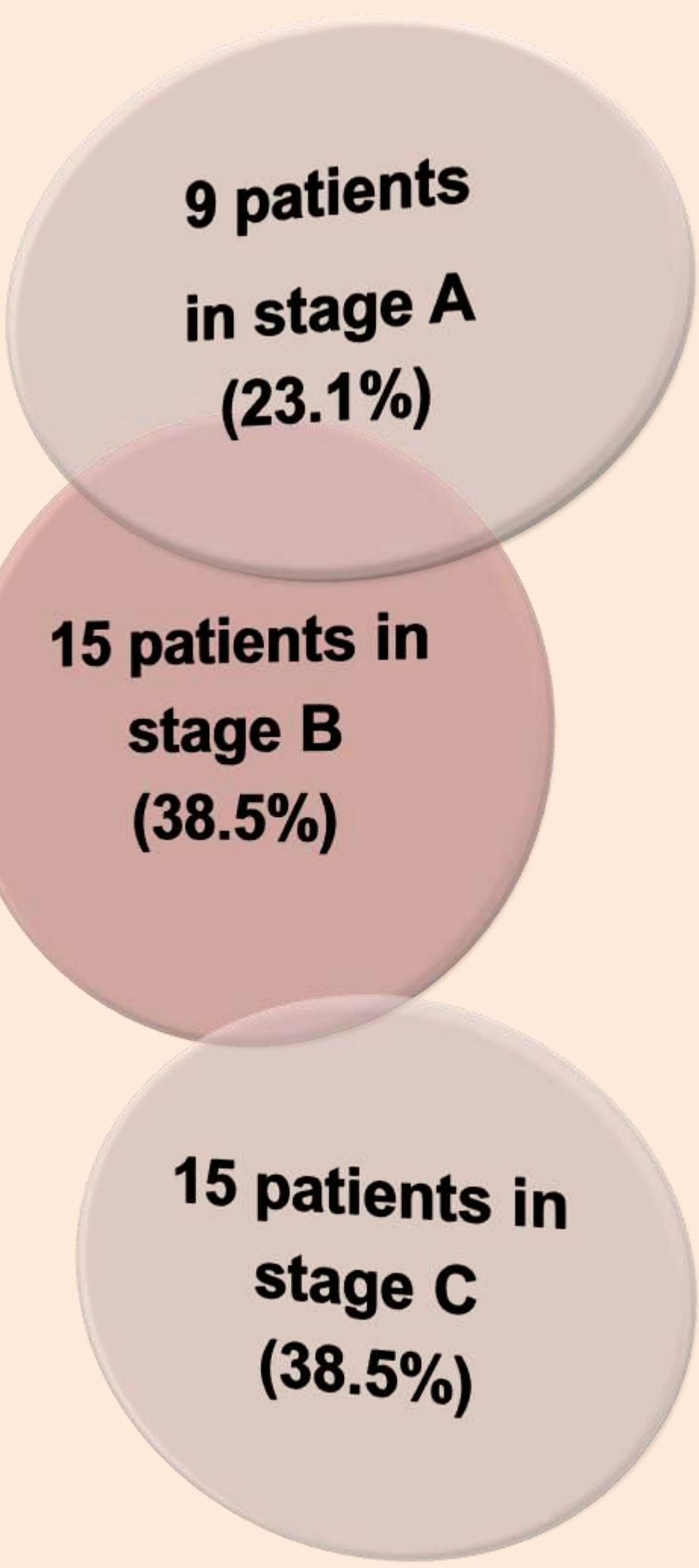
Analysis and statistical study

The expression of ZAP-70 was determined by flow cytometry on a Beckman Colter DxFlex cytometer at the hemobiology-blood transfusion department of the Constantine University Hospital. For the statistical study, we used Fisher's exact test for the correlation between ZAP-70 expression and the other criteria, namely the Binet prognostic classification, the treatment-free survival duration, and the disease progression.

RESULTS & DISCUSSION



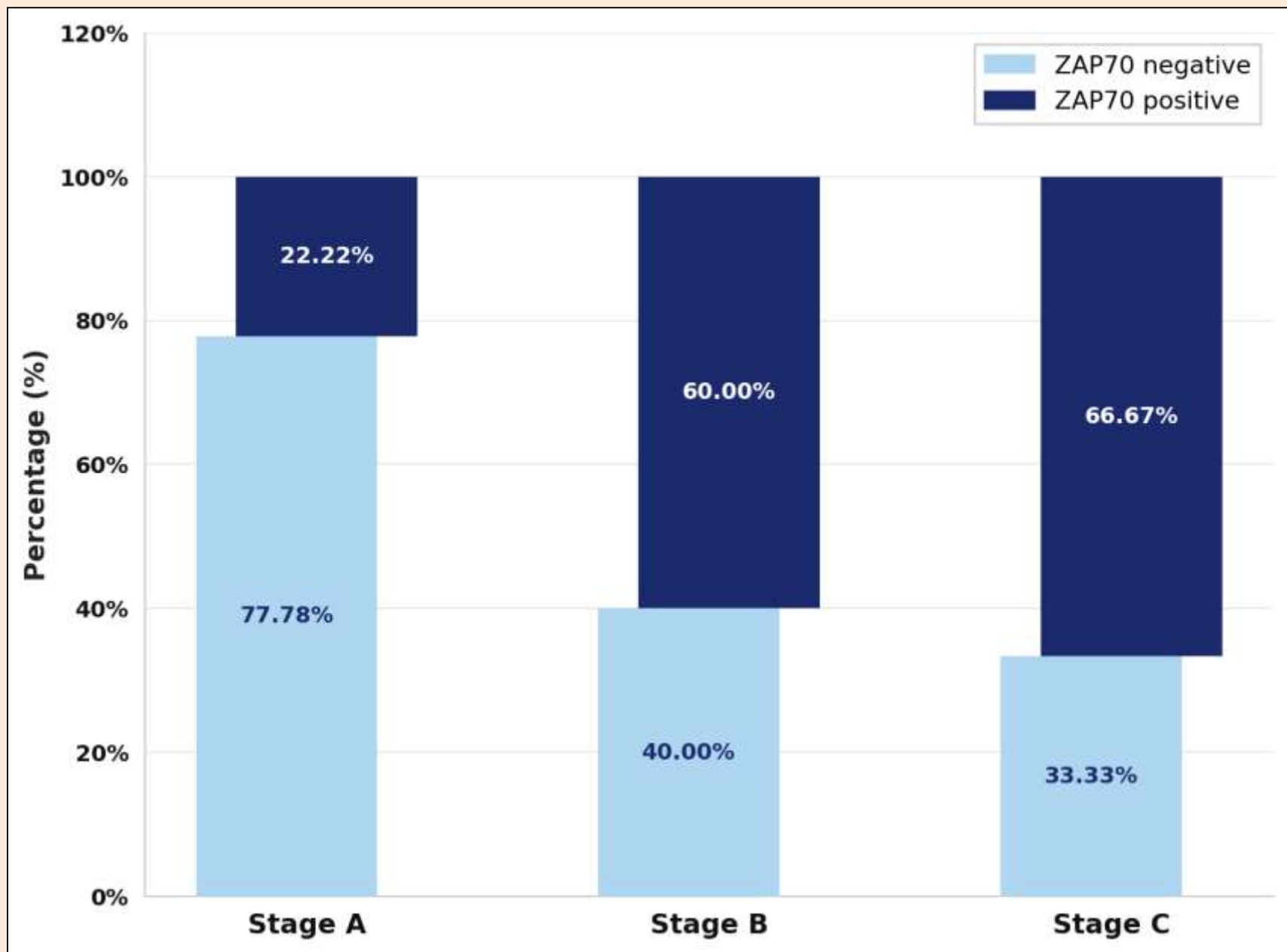
Sexe ratio :
H/F : 2/1



21 (53.8%)
expressed ZAP-70 at a rate $\geq 20\%$ and considered ZAP-70 positive

8 patients (46.2%)
exhibited an expression below 20% and were considered ZAP-70 negative

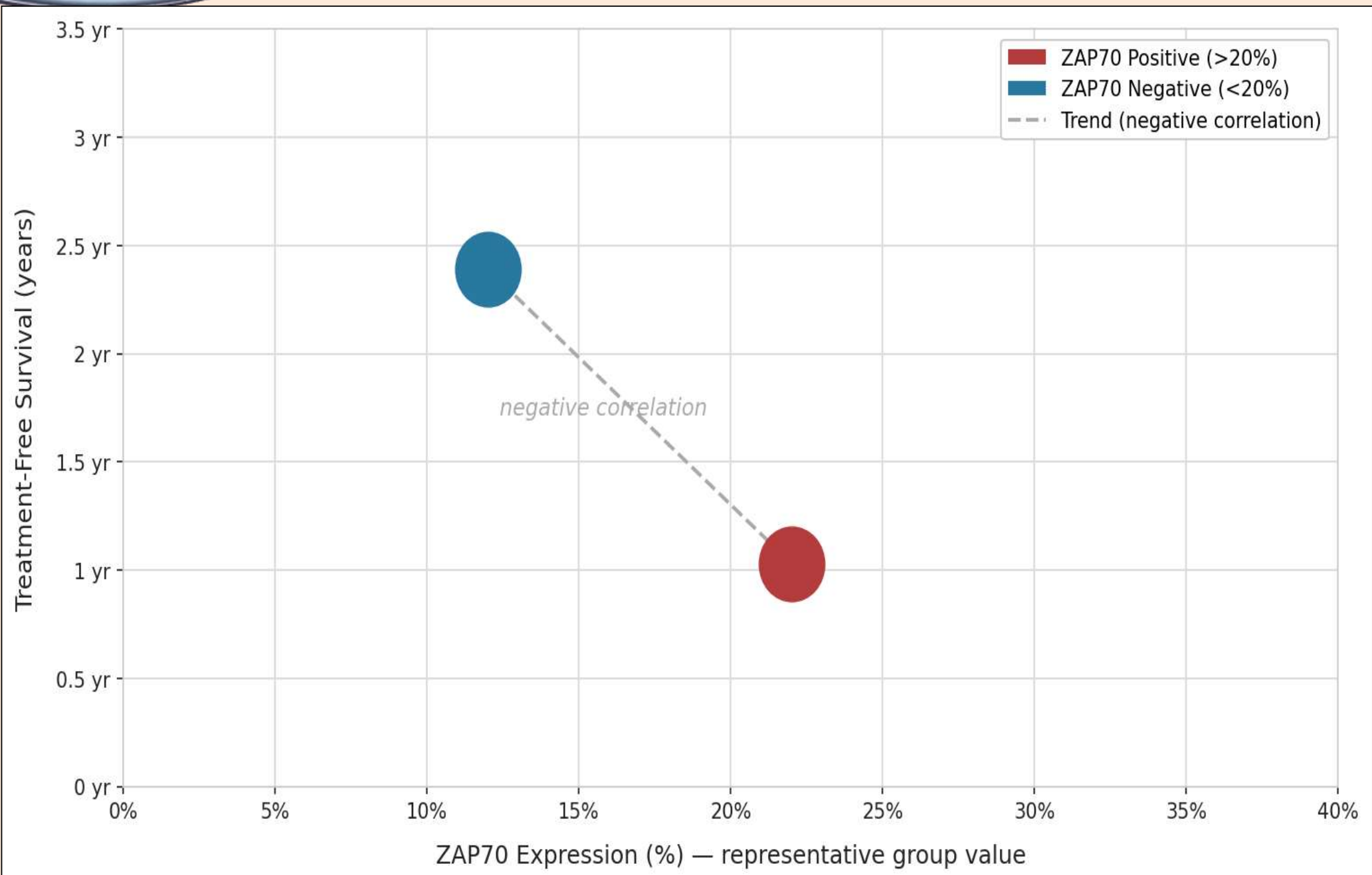
The clinical progression was good in 56.4% of our patients, while 43.6% showed poor progression, taking into account several clinico-biological criteria for assessing this progression.



Distribution of ZAP-70 status according to Binet stage

ZAP-70 Status	Favourable Outcome	Unfavourable Outcome	Total (%)
ZAP-70 Positive	38.1 %	61.9%	53.8%
ZAP-70 Negative	77.8%	22.2 %	46.2%
Total	56.4%	43.6%	100 %

Correlation between ZAP-70 expression and disease outcome



Correlation between ZAP-70 expression and treatment-free survival

CONCLUSION

ZAP-70 is a clinically relevant prognostic marker in CLL, associated with unfavorable clinical evolution and shortened treatment-free survival. Its systematic assessment at diagnosis is strongly recommended to improve risk stratification in CLL patients.